Chapter 6 Part 3: Patterns of Inheritance

- The genetic secrets that Mendel discovered
- Can we predict the chances of passing on certain genes to offspring?
- Sex-linked genes
- 5 realms of non-Mendelian genetics
- Genetic anomalies
- The environment affects traits too

Corresponds with OpenStax Biology 2e Chapter 12 & 13

The genetic secrets that Mendel discovered

- In the 1800s, no one knew anything about inheritance – only that offspring resembled their parents
  – Gregor Mendel, an Austrian monk, discovered some of the most basic principles of genetics
    • used pea plants in all of his experiments

Gregor Mendel (1822–1884)

Figure: https://commons.wikimedia.org/wiki/File:Gregor_mendel.jpg
Figure: https://www.pxfuel.com/en/free-photo-jhczc/download

- Why use pea plants?
  - They are easy to maintain & breed
  - They reproduce quickly, so multiple generations can be observed
  - They have many easy to see traits with only two variants each

Easy to see traits, only 2 variants each

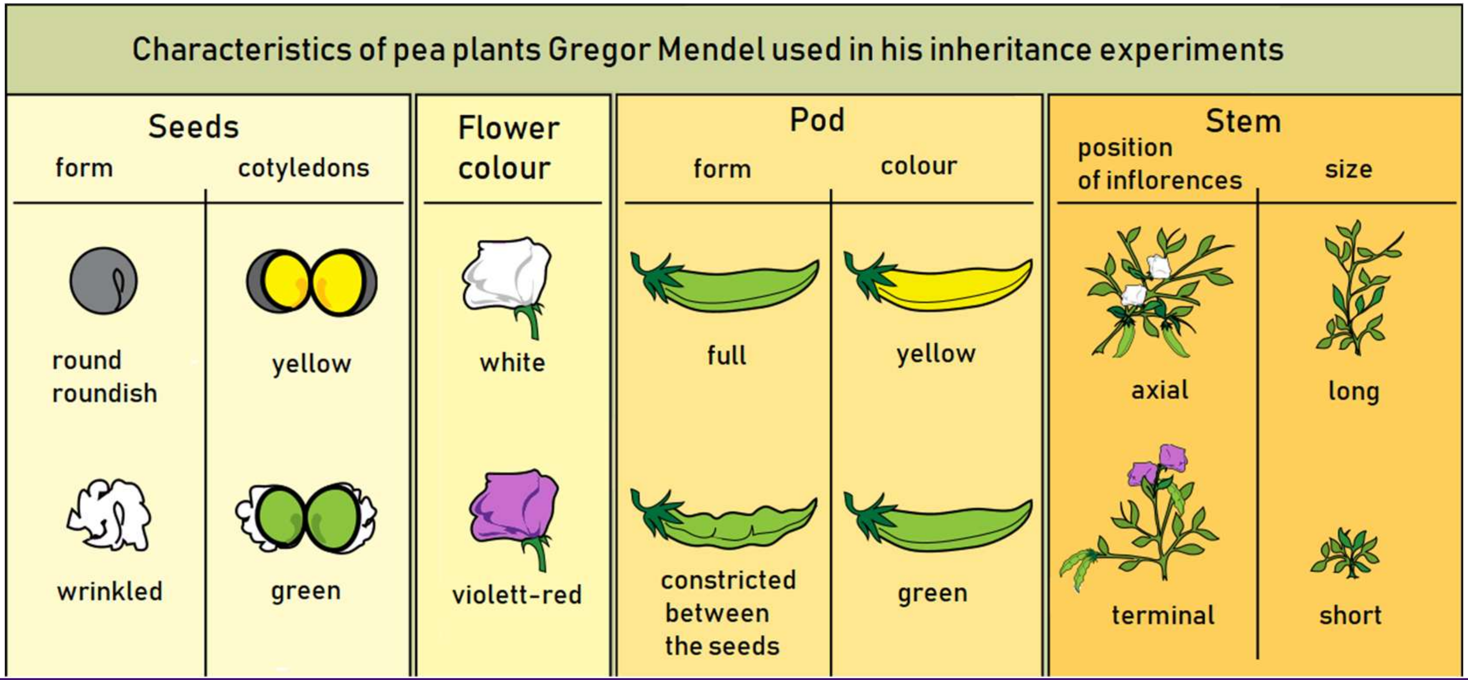


Figure: https://commons.wikimedia.org/wiki/File:Gregor_Mendel_-_characteristics_of_pea_plants_-_english.png

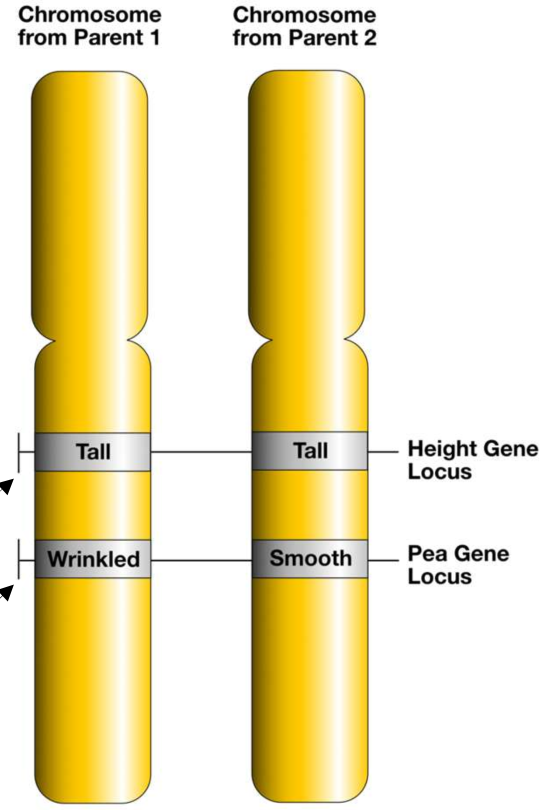
- What did Mendel discover?
  - *Remember*: we knew that offspring resemble parents → "heredity"
    - So something must be passed on from parents to offspring, but what?

- Mendel's work uncovered the secrets of "**genes**"
  - definition for gene then was simple: a "unit of information" passed on to offspring

    (we didn't know DNA even existed, so we definitely didn't know it made up genes or was involved with inheritance)

Figure: https://www.piqsels.com/en/public-domain-photo-jkdrq

- <u>1. All organisms have 2 of every gene</u>
  - Through very careful notes on thousands of pea plant crosses, Mendel discovered that all organisms have **2 of every gene**
    - 1 from each parent
      - *we now know they are on homologous chromosomes*



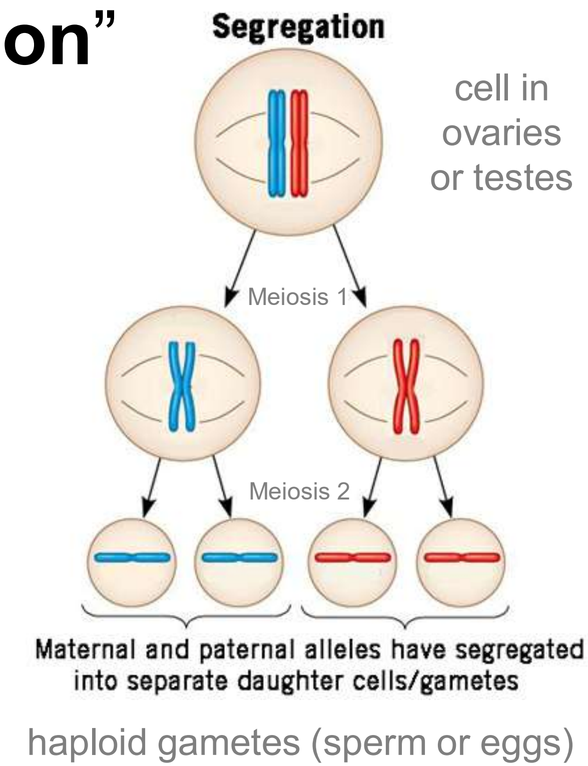
2 genes in a plant with the same allele to make it tall
2 genes in a plant with different alleles, one for wrinkled & one for smooth peas

Figure: https://commons.wikimedia.org/wiki/File:Gene_Loci_and_Alleles.png

- <u>2. Only 1 of an organism’s 2 genes is passed on to offspring</u>
  - – Even though a parent has 2 of every gene, they only pass 1 on to their offspring
  - – Also called the “**Law of segregation**”

    = each gene is separated during gamete formation, so that gametes only have 1 gene or the other

    - *We now know this occurs because of meiosis*



- 3. The 1 gene that is passed on is random – 50/50 chance
  - 50% of gametes carry 1 of the genes
  - 50% of gametes carry the other gene
    - There is a 50/50 chance that a parent will pass on 1 gene or the other

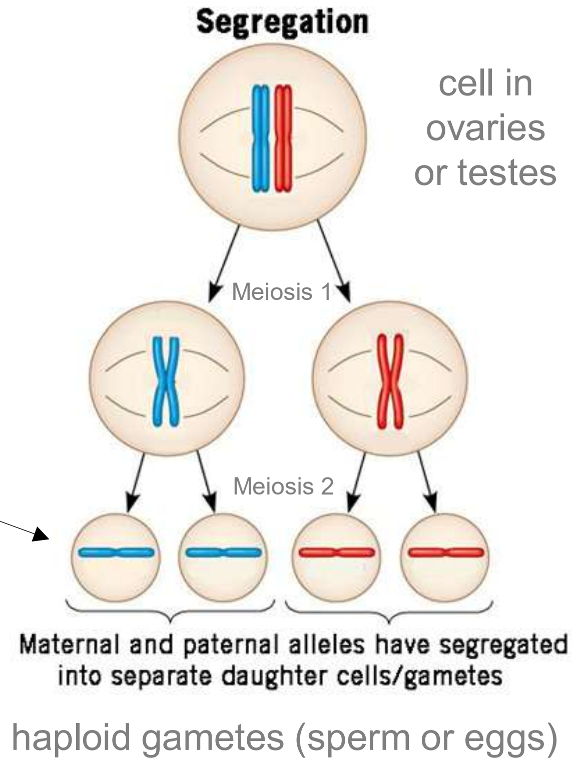
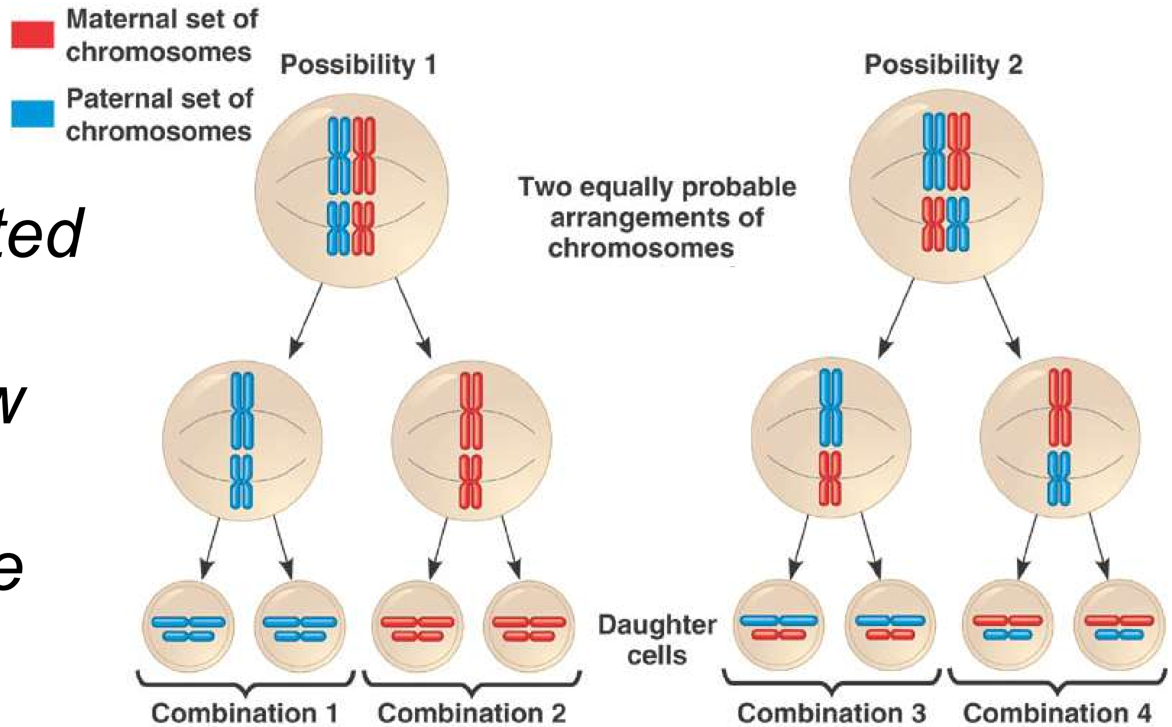


Figure: http://www.bio.miami.edu/dana/250/250S19_2.html

- <u>3. The 1 gene that is passed on is random – 50/50 chance</u> *(cont.)*
  - Also part of the "**law of independent assortment**"
    - Because it is independent of the other genes passed on
  - *Genes are separated from each other & don't influence how other genes are separated – they're all 50/50*



- 4. An organisms genes can be homozygous or heterozygous
  - Since organisms have 2 copies of every gene, it is possible to have:
    - 2 matching alleles = **homozygous** for that gene
      - e.g. plant height for tall
    - 2 different alleles = **heterozygous** for that gene
      - e.g. pea shape for wrinkled AND smooth

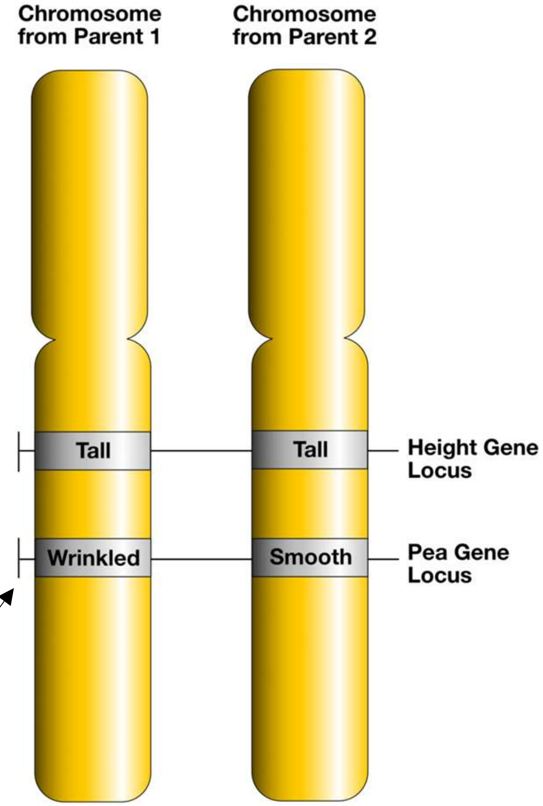
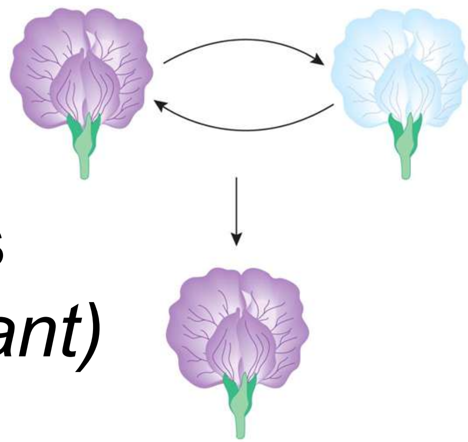


Figure: https://commons.wikimedia.org/wiki/File:Gene_Loci_and_Alleles.png

- <u>5. Some alleles are dominant or recessive</u>
  - If an organism is *homozygous* for a gene, they will only show that 1 possibility
    - e.g. purple flower allele + purple flower allele = having purple flowers
  - BUT: in a *heterozygous* case, **the dominant allele will mask the recessive allele**
    - e.g. purple flower allele + white flower allele = having purple flowers *(because the purple allele is dominant)*



- <u>5. Some alleles are dominant or recessive</u> *(cont.)*
  - We usually denote dominant alleles with capital letters & recessive alleles with lowercase letters
    - Purple allele = dominant = “P”
    - White allele = recessive = “p”
  - *Homozygous dominant = PP*
  - *Heterozygous = Pp*
  - *Homozygous recessive = pp*

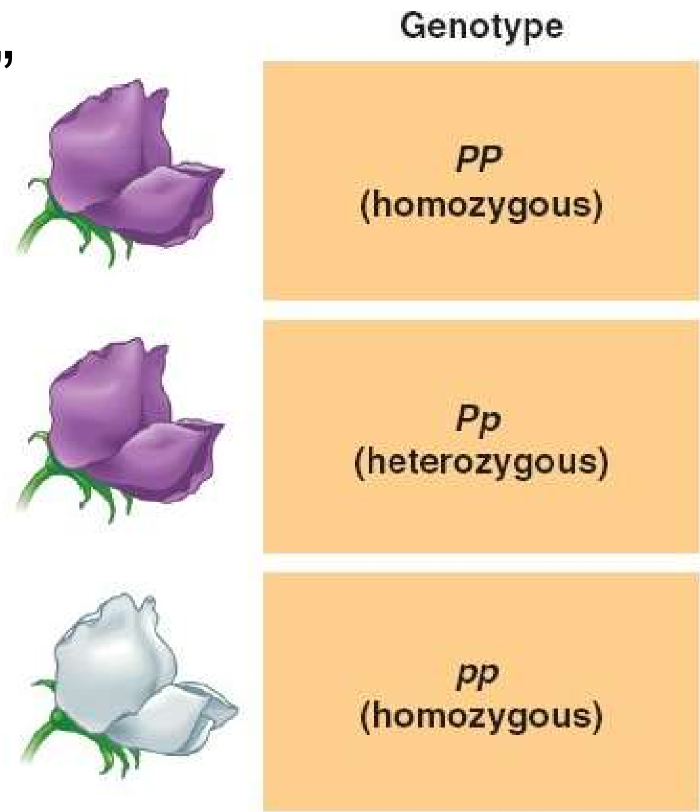


Figure: http://cikgurozaini.blogspot.com/2010/06/genetic-1.html

- Since a dominant allele can mask a recessive allele, we now see that <u>the genes we have inside our cells don’t always match the traits we see</u>
  - The 2 alleles carried by an organism = the **genotype** *(what you <u>have</u>)*
    - *e.g. purple & white alleles, or “Pp”*
  - The physical expression of the genotype = the **phenotype** *(what you <u>show</u>)*
    - *e.g. purple flowers*

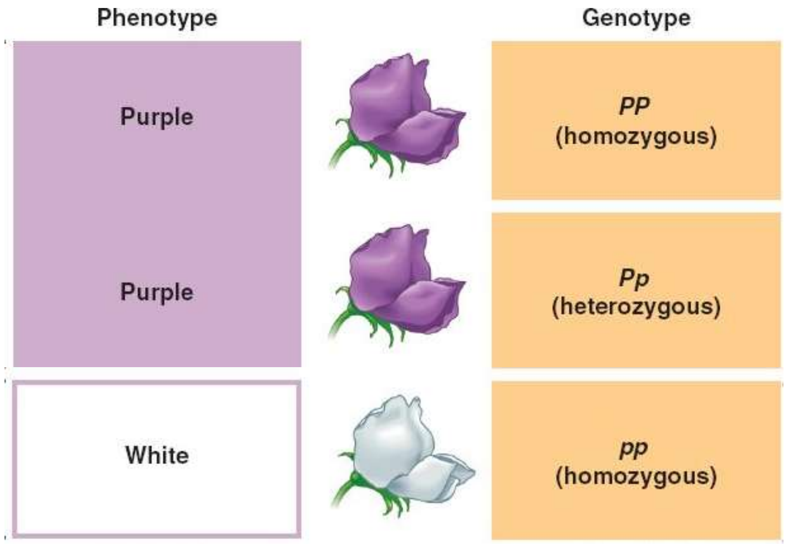
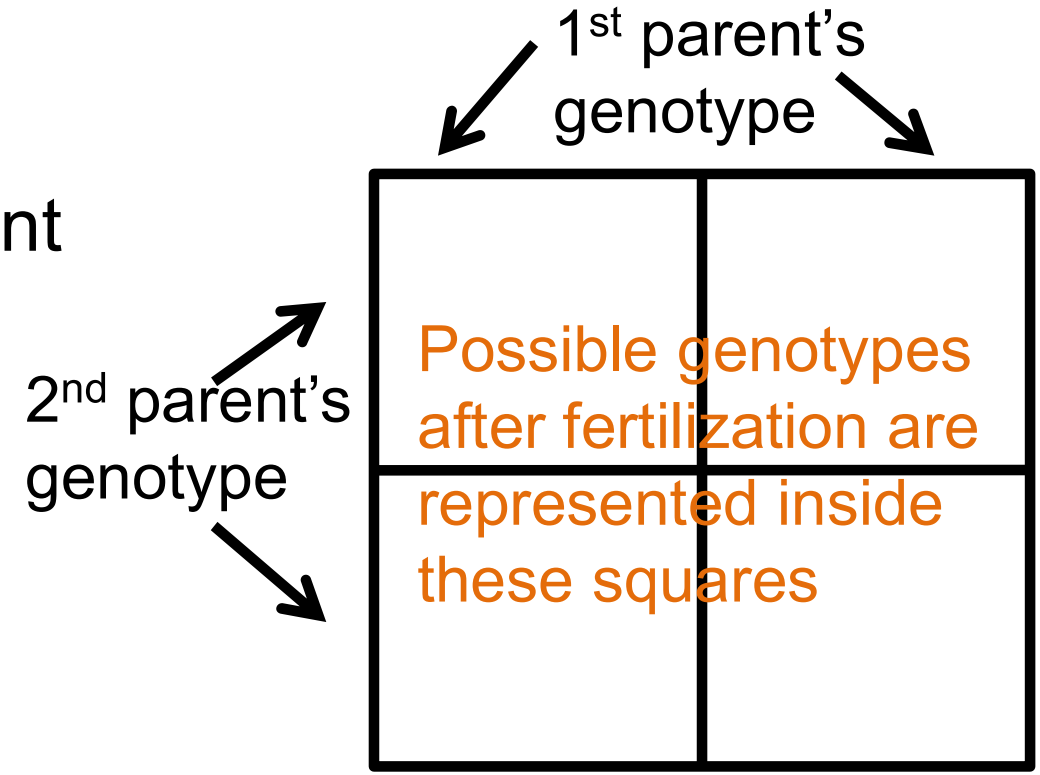


Figure: http://cikgurozaini.blogspot.com/2010/06/genetic-1.html

Can we predict the chances of passing on certain genes to offspring?

- The **Punnett square** method predicts the probability of offspring genotypes & phenotypes using the genotype of each parent



- Let’s try it for the flower color trait:
  - 1 parent has genotype Pp (has purple flowers)
  - Other has genotype pp (has white flowers)

One parent is Pp

One parent is pp

	P	p
p	Pp	pp
p	Pp	pp

- <u>What is the % chance of each **genotype**?</u>
  - Homozygous dominant (PP) = 0%
  - Heterozygous (Pp) = 50%
  - Homozygous recessive (pp) = 50%

	P	**p**
p	**Pp**	**pp**
p	**Pp**	**pp**

- **What is the % chance of each phenotype?**
  - Purple-flowered offspring = 50%
  - White-flowered = 50%

	P	p
p	Pp	pp
p	Pp	pp

Figure: https://commons.wikimedia.org/wiki/File:White_butterfly_pea.jpg
Figure: https://www.flickr.com/photos/23271361@N06/8214598104

- <u>Let’s try it with another trait:</u>
  - The gene for fur length in cats has 2 alleles:
    - Short fur = dominant (F)
    - Long fur = recessive (f)
  - 1 parent is homozygous dominant with short fur (FF)

  - 1 parent is homozygous recessive with long fur (ff)

  - What would the genotypes & phenotypes of their offspring be regarding fur?

Figure: https://www.piqsels.com/en/public-domain-photo-ffoxk
Figure: https://pixnio.com/fauna-animals/cats-and-kittens/yellow-cat-animal-outdoor-yellow-grass-nature-eye-fur-whisker

- <u>What is the % chance of each **genotype**?</u>
  - Homozygous dominant =
  - Heterozygous =
  - Homozygous recessive =

One parent is FF

One parent is ff

Figure: https://www.piqsels.com/en/public-domain-photo-ffoxk
Figure: https://pixnio.com/fauna-animals/cats-and-kittens/yellow-cat-animal-outdoor-yellow-grass-nature-eye-fur-whisker

- <u>What is the % chance of each **phenotype**?</u>
  - Short fur =
  - Long fur =

One parent is FF

One parent is ff

Figure: https://www.piqsels.com/en/public-domain-photo-ffoxk
Figure: https://pixnio.com/fauna-animals/cats-and-kittens/yellow-cat-animal-outdoor-yellow-grass-nature-eye-fur-whisker

- <u>NEW PRACTICE</u>
  - 1 parent is heterozygous with short fur

  - The other parent is also heterozygous with short fur

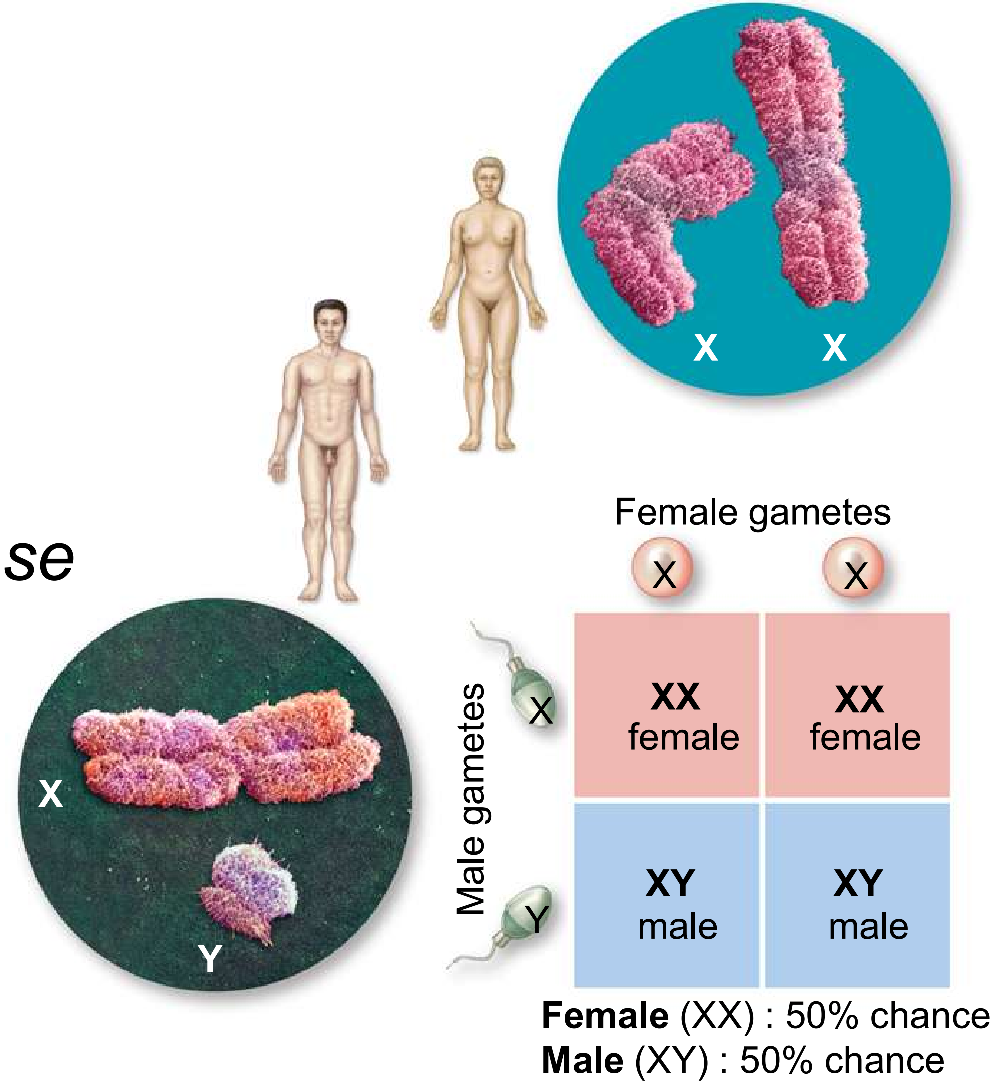
- <u>What is the % chance of each **genotype**?</u>
  - Homozygous dominant =
  - Heterozygous =
  - Homozygous recessive =

Figure: https://commons.wikimedia.org/wiki/File:Cat_March_2010-1.jpg
Figure: https://pixabay.com/photos/cat-kitten-tabby-cat-pet-5564211/

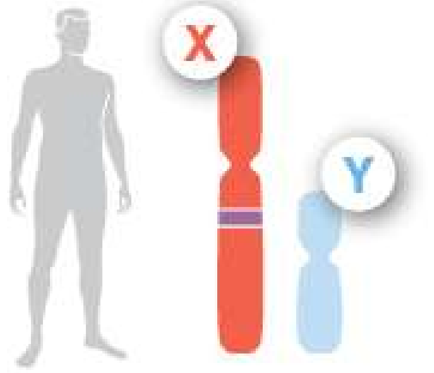
- <u>What is the % chance of each **phenotype**?</u>
  - Short fur =
  - Long fur =

Sex-linked genes

- <u>*Remember*: sex is determined by sex chromosomes</u>
  - *Also remember: not everyone fits into these 2 categories*
  - *Also remember: sex is not gender*



- **Sex-linked genes** are found only on the X or on the Y sex chromosomes (not both)
  - Some genes found only on the X chromosome are important to both sexes
    - e.g. genes for color vision, blood clotting, & certain proteins in muscles

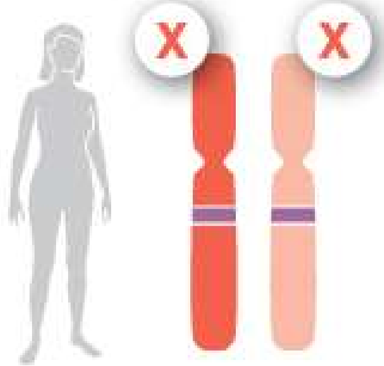


Males will have only 1 of these important genes, inherited from his female parent

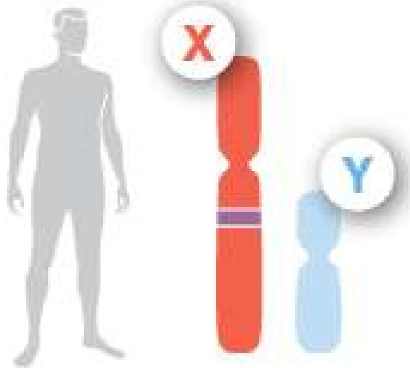
Females will have 2 of these important genes, 1 from each parent

Males only have one copy of the X chromosome

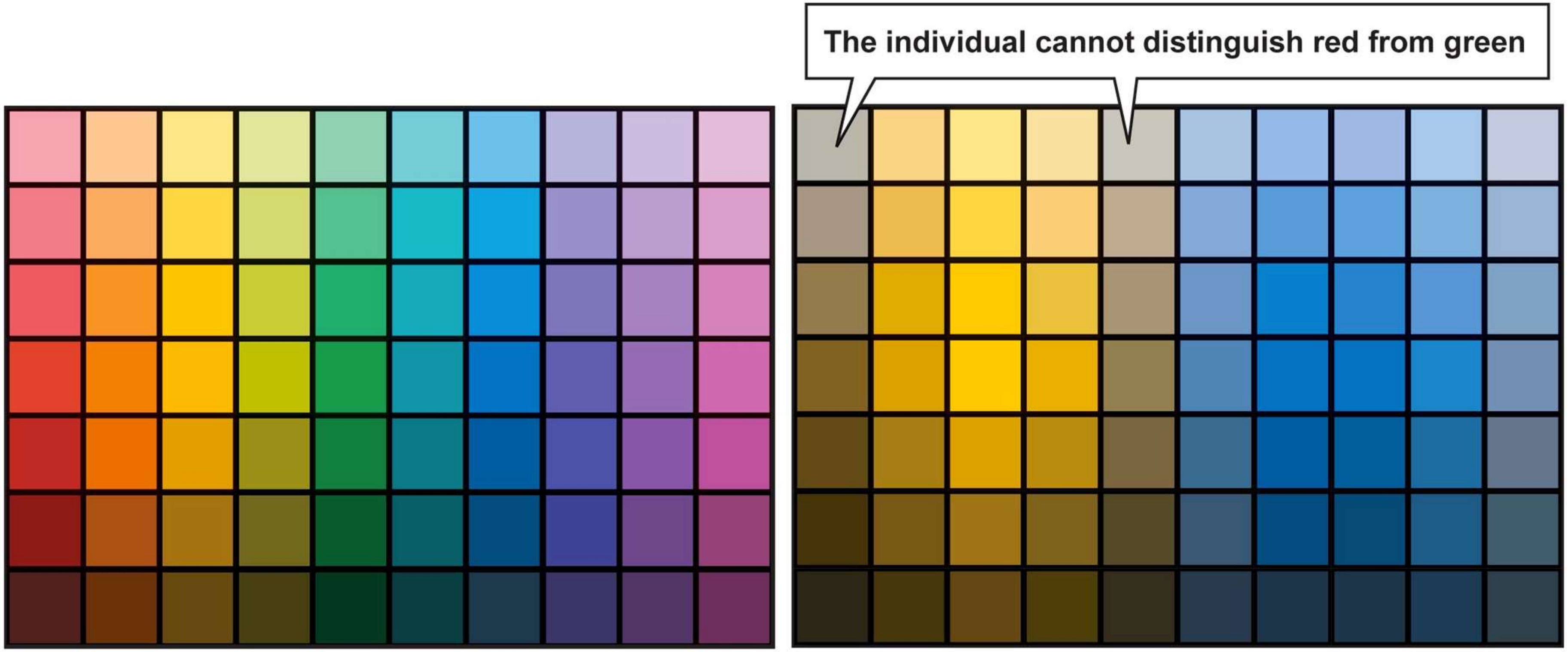
- Females (XX) can be homozygous or heterozygous for a characteristic
  - If heterozygous: the dominant allele shows up



- Males (XY) have only 1 copy of each gene - there is no 2nd copy to mask recessive genes
  - Fully express all the X-linked genes inherited from their female parent, whether the alleles are dominant or recessive



- A gene on the X-chromosome codes for the protein in the eye that detects red & green = *sex-linked gene*
  - Red-green color blindness in humans is caused by a recessive allele on the X chromosome that does not properly code for the protein



(a) Normal color vision **(b) Red-green color blindness**

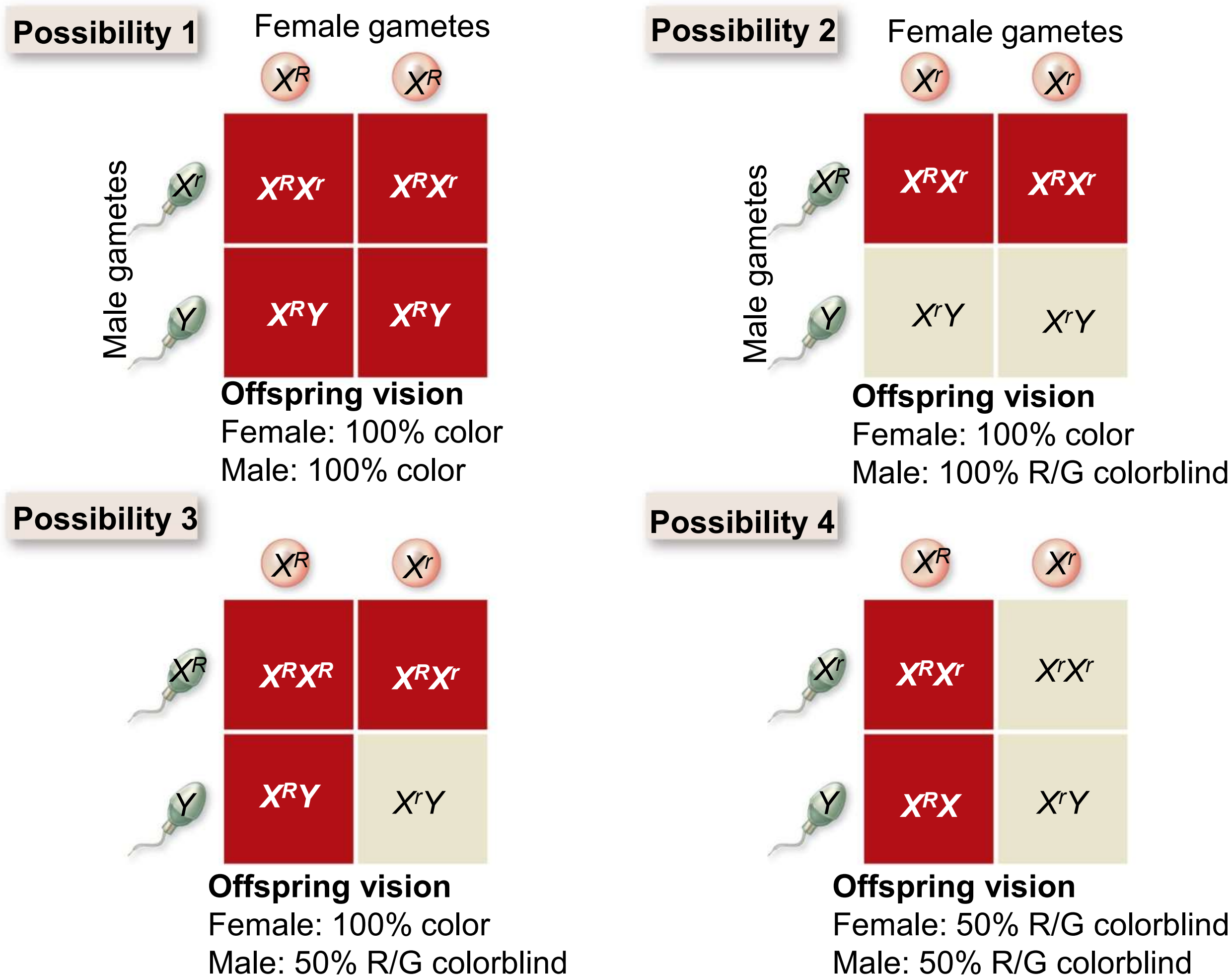
- <u>Inheriting sex-linked genes</u>
  - X_R = dominant allele = color vision
  - X_r = recessive allele = R/G color blind
  - **Females** can have [X_R-X_R], or [X_R-X_r], or [X_r-X_r]
    - Only the [X_r-X_r] combination = R/G color blind
      - *A female with R/G colorblindness inherited it from BOTH parents*

        The male parent must also be R/G color-blind

- But **males** have only 1 of every gene on the X chromosome
  - If they inherit [X_R-Y], then they have normal vision
  - If they inherit [X_r-Y], then they are R/G color blind
    - There is no option for a dominant allele (X_R) to mask a recessive one
    - This means <u>males = 16 times more likely to be color blind than females</u>
      - *A male with R/G colorblindness inherited it from his female parent ONLY – the male parent can have regular color vision*

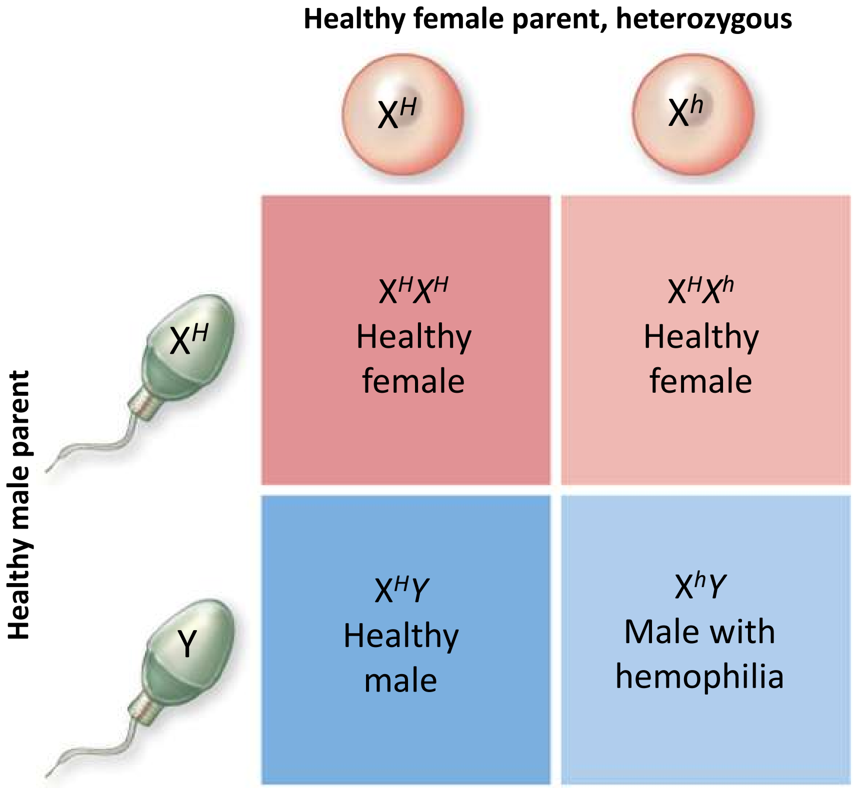
- <u>Possible parental combinations & resulting offspring</u>
  - – Notice how much less likely it is to produce a female that is R/G colorblind

There are more possible combinations - here are only 4



- <u>Other sex-linked genes (on X chromosome), which affect males 16x more, include:</u>
  - **Muscular dystrophy** (progressive weakening of the muscles)
  - **Pattern Baldness** (most common type of baldness)
    - "pattern" because it recedes from the hairline 1st, then becomes thinner, then is only left on back & sides of head

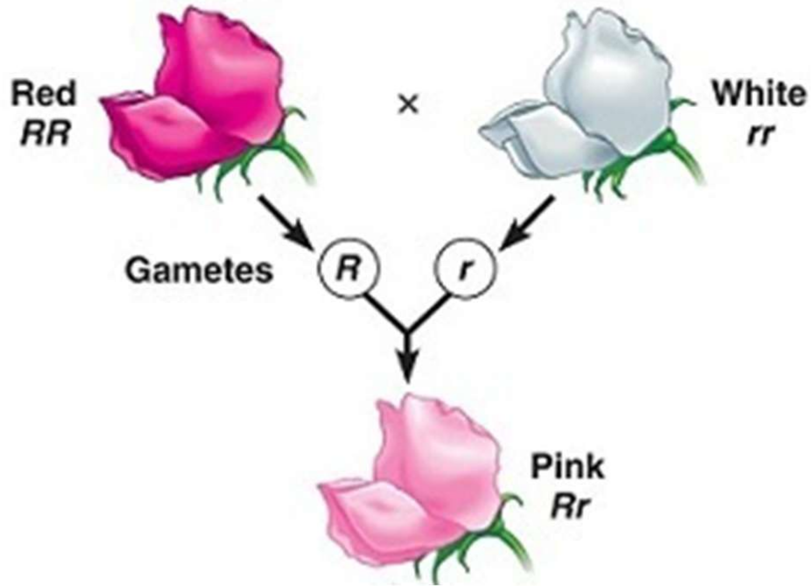
- <u>Other sex-linked genes (on X chromosome), which affect males 16x more, include:</u>
  - **Hemophilia** (in which the ability of the blood to clot is severely reduced)



5 realms of non-Mendelian genetics

- Many traits do not follow the simple Mendelian rules of inheritance
  - There are 5 realms where genetics are a bit more complex: **non-Mendelian genetics**
    - 1. Incomplete dominance
    - 2. Multiple alleles
    - 3. Codominance
    - 4. Polygenic inheritance
    - 5. Pleiotropy

- 1. Some alleles are not dominant & recessive, but are *incompletely* dominant over others
  - When the heterozygous phenotype is an intermediate between the 2 homozygous phenotypes, the pattern of inheritance is called **incomplete dominance**



e.g. RR = red rose color

WW = white rose color

RW = pink rose color

Incomplete dominance results in an *in-between* phenotype

- Another example of incomplete dominance:

e.g. CC = curly hair

SS = straight hair

<u>CS = wavy hair</u>

Notice that we don't use the same letter capitalized & lowercase, because neither is dominant over the other

Incomplete dominance results in an *in-between* phenotype

Figures from https://www.wig.com/

- 2. A single gene may have **multiple alleles**
  - The human blood types are an example of multiple alleles of a single gene
    - Human blood type genes have 3 possible alleles: A, B, and O
  - Consequences of multiple alleles:
    - instead of having only 3 possible genotypes (e.g. RR, Rr, rr), there are many possible genotypes:
      - AA
      - AB
      - AO
      - BB
      - BO
      - OO

- 3. Some alleles are **codominant**: both are expressed in entirety
  - Human blood types are another good example of this: A & B are codominant over O, which is recessive
    - leads to an extra phenotype: blood type AB

Genotype (2 alleles)	Phenotype (blood type)	Description
AA	A	homozygous
BB	B	homozygous
OO	O	homozygous
AO	A	heterozygous – A is dominant
BO	B	heterozygous – B is dominant
AB	AB	heterozygous – codominance

- What does blood type mean?
  - <u>Blood type refers to the antigens a cell makes</u>
    - Antigens = recognition proteins that sit on the outside of a cell & prevent the immune system from attacking them

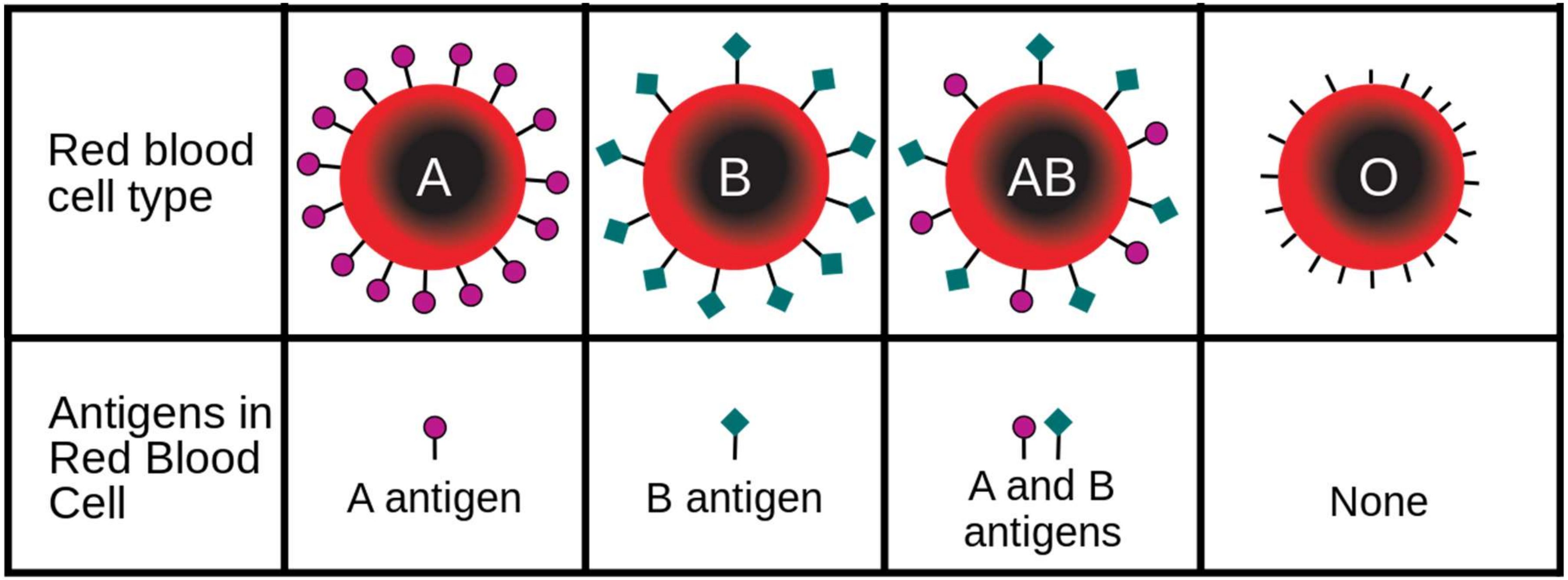


Figure: https://en.wikipedia.org/wiki/ABO_blood_group_system

- <u>We make antibodies against antigens we don't have</u>
  – AB has both antigens, so makes no antibodies
  – O has no antigens, so makes both antibodies

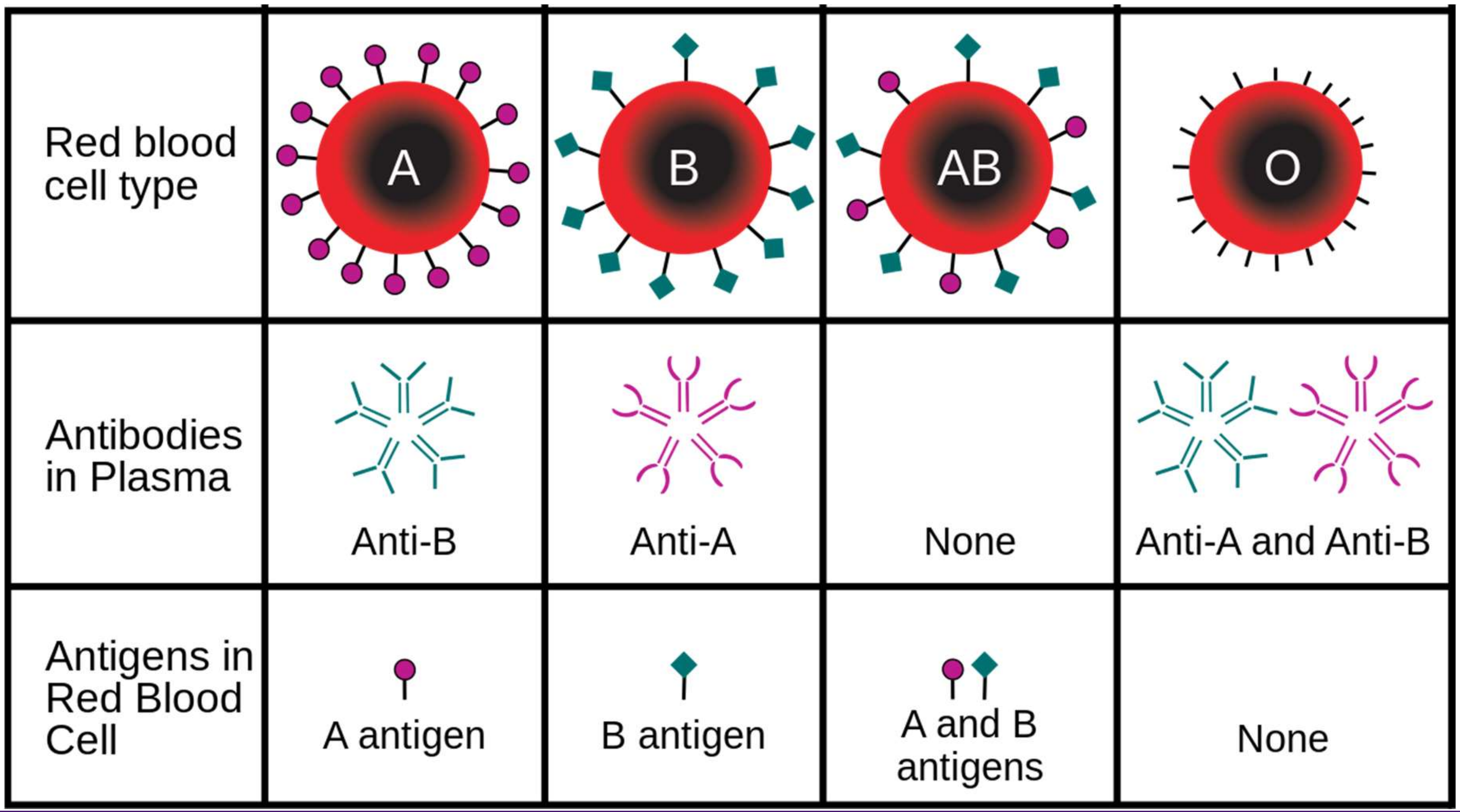


Figure: https://en.wikipedia.org/wiki/ABO_blood_group_system

- <u>What does this mean for blood donation?</u>
  - Type O blood is not attacked by anyone: **universal donor**
  - Type AB blood has no antibodies, so will not attack any kind of blood: **universal recipient**

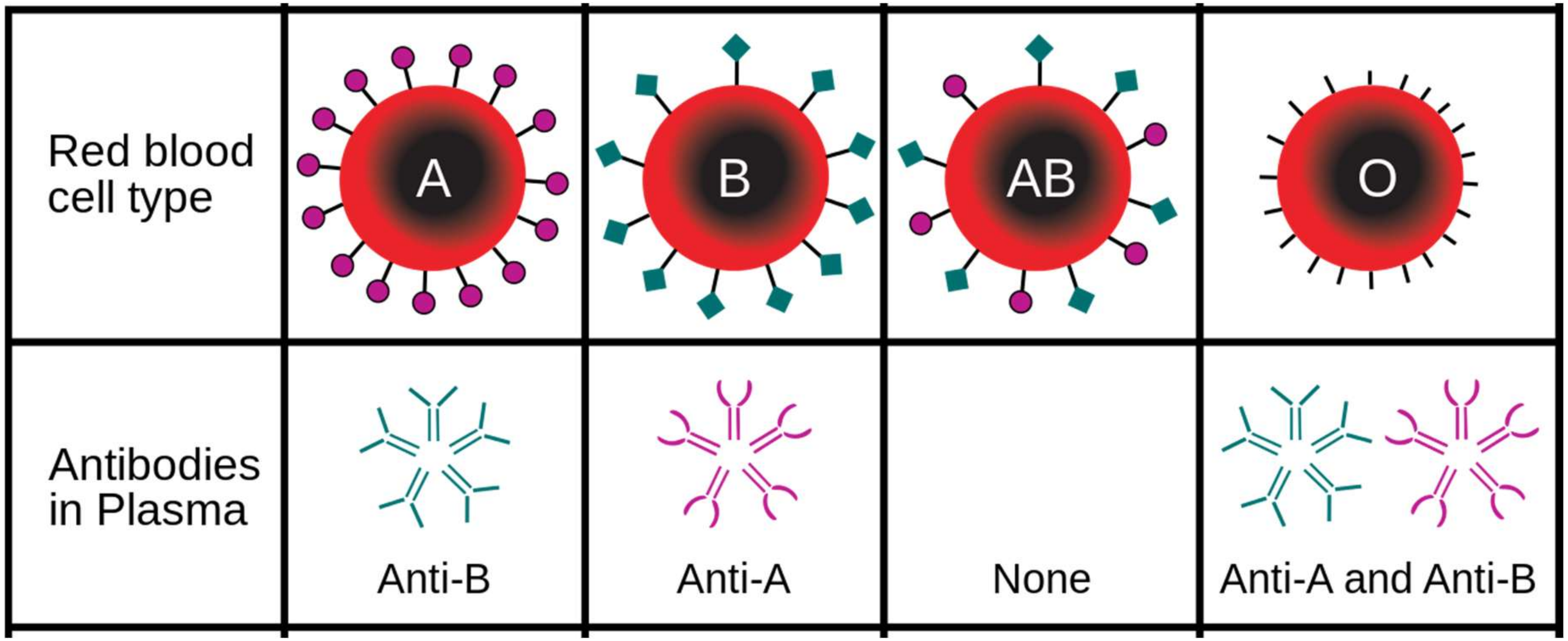


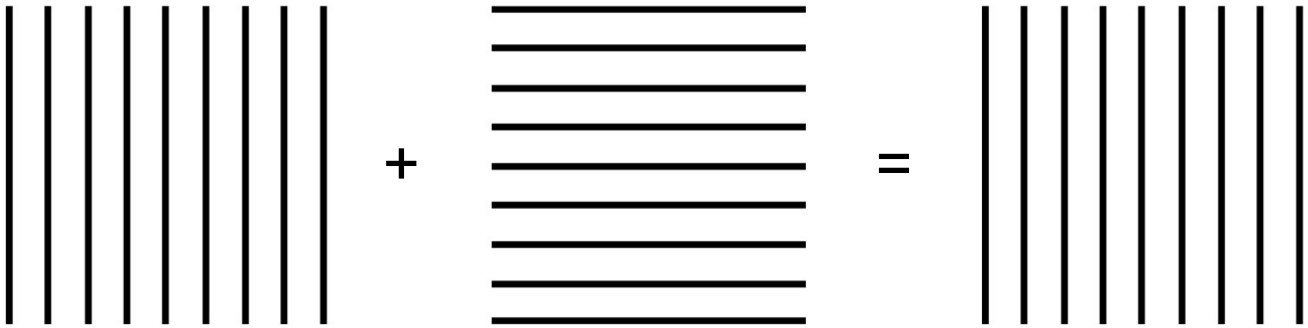
Figure: https://en.wikipedia.org/wiki/ABO_blood_group_system

- <u>Phenotypes & genotypes in US population</u>
  - Type O blood makes antibodies for both A & B, so they can ONLY get blood transfusions with other type O blood
    - Fortunately, almost half the US population has type O blood

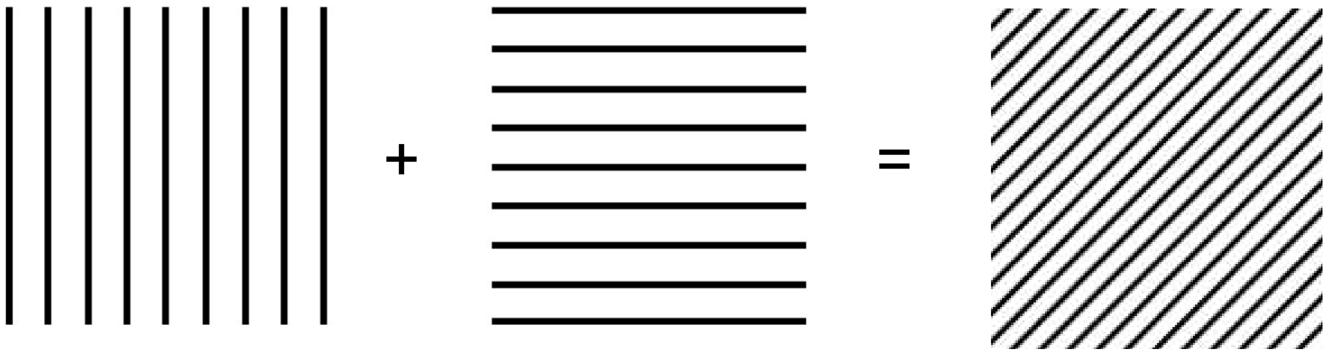
"Blood type" Phenotype	Genotypes	Antibodies in the blood	% of US pop. w/blood type
Type A	AA and AO	Anti-B only	40%
Type B	BB and BO	Anti-A only	10%
Type AB	AB	None	4%
Type O	OO	Anti-A and Anti-B	46%

Understanding the 3 possibilities of dominance

Dominant & recessive:



Incomplete dominance:



Codominance:

- **4. Polygenic inheritance** *"polygenic" = many genes*
  - Traits based on 1 gene have specific phenotypes
    - e.g. short fur or long fur
    - e.g. blood type A, B, AB, or O
  - But some traits don't have specific phenotypes – they have a range
    - e.g. height doesn't come only in 5'5", 5'10", & 6'2"
      - it comes in a range from 4' all the way up to 7' or more

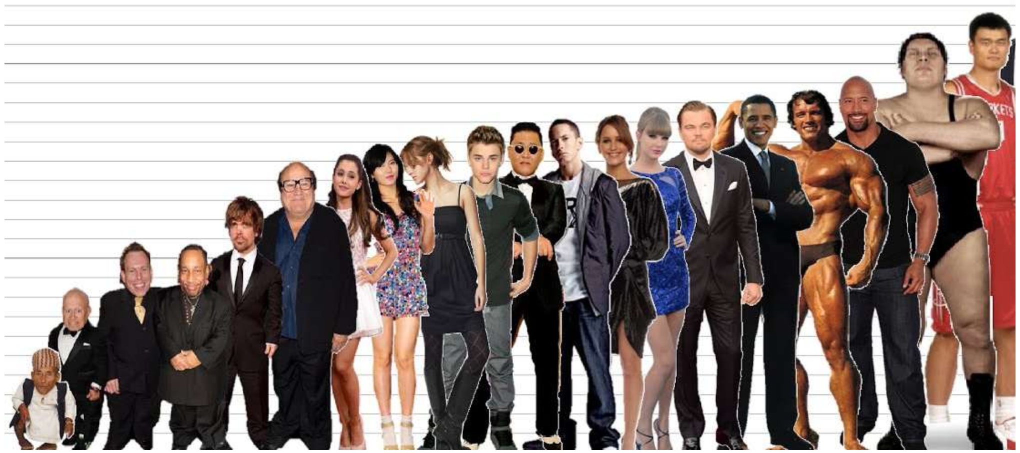


Figure: https://www.youtube.com/watch?v=E0Aeks_id6c

- Phenotypes that have a range (instead of specific phenotypes) are caused by multiple genes: **polygenic inheritance**
  - Human examples = height, skin color, eye color, & body build
    - Skin color is due to at least 3 different genes
      - *many genes, one trait*

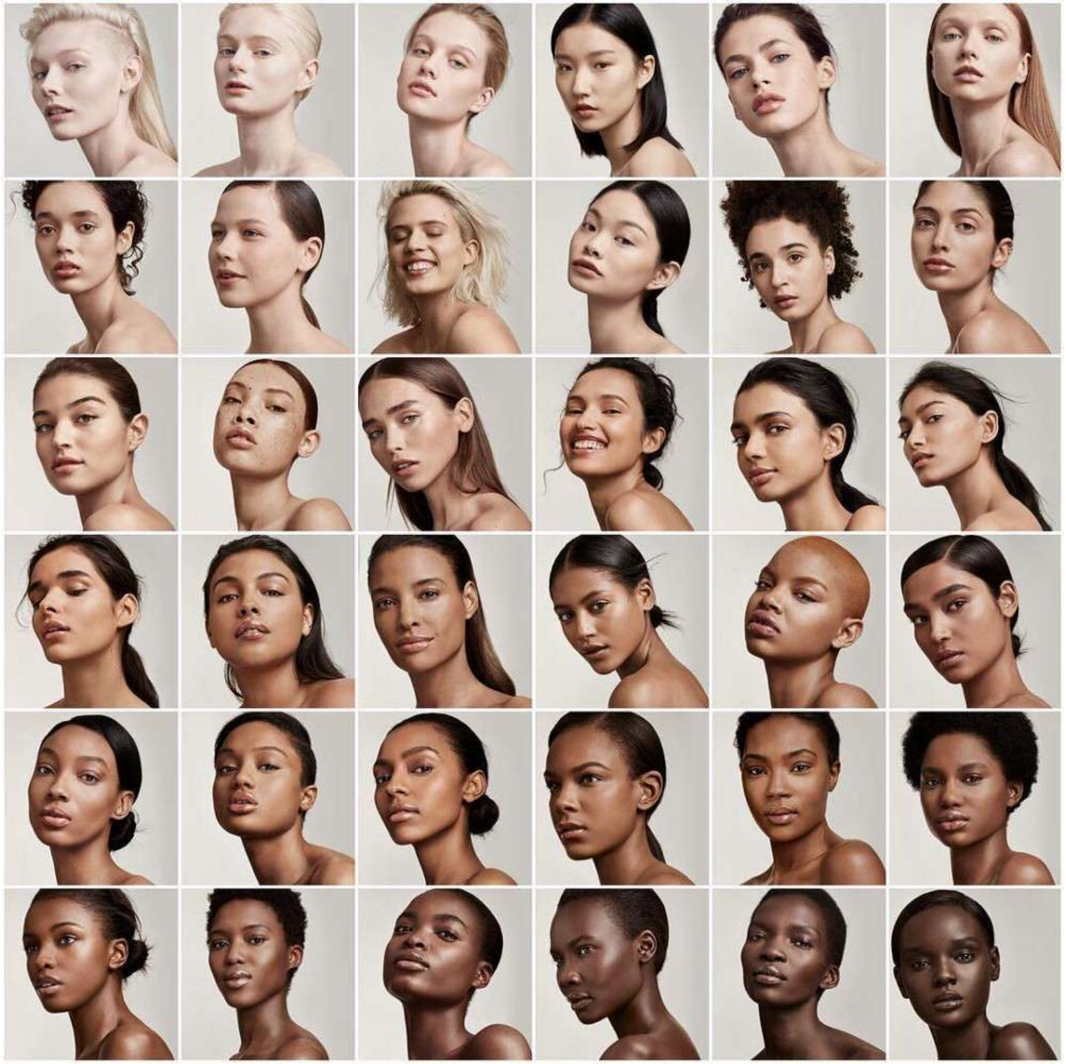


Figure: https://fentybeauty.com/

- 5. Sometimes 1 gene doesn't only have 1 trait – sometimes 1 gene affects many traits = **pleiotropy**
  - e.g. in chickens, a gene called "frizzle" affects:
    - Feather pattern
    - Metabolic rate
    - Body temperature
    - Digestive ability
  - *one gene, many traits*
    - *opposite of polygenic*

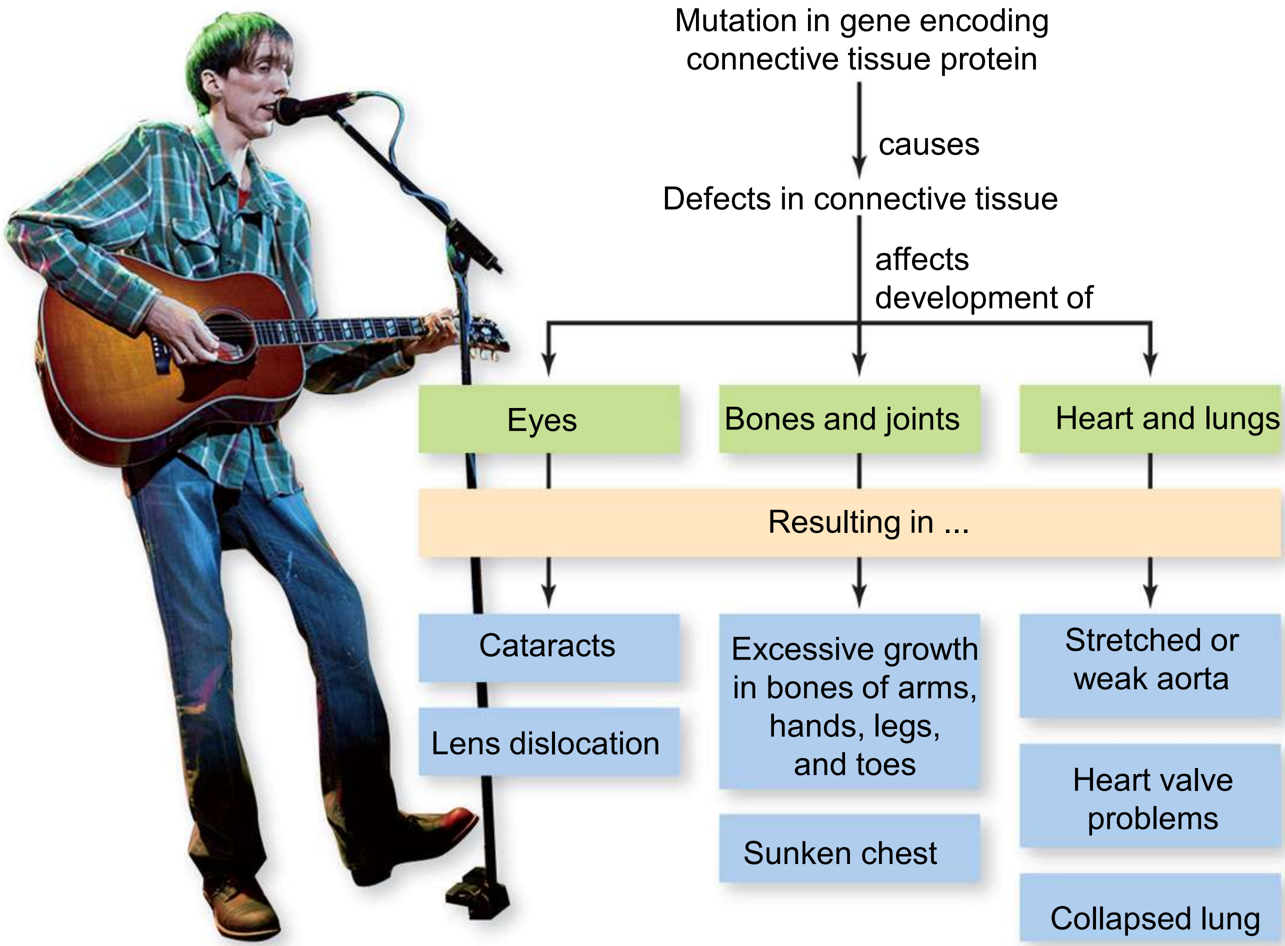
Figure: https://commons.wikimedia.org/wiki/File:Leghorn_Frizzle_Chicken.jpg

- A human example of pleiotropy = Marfan syndrome
  - *A mutated gene on chromosome 15 causes an error in coding for a protein important for connective tissue*

Bradford Cox of the band Deerhunter has Marfan syndrome

Abraham Lincoln is suspected to have had it too

One gene, affecting many traits



Genetic anomalies

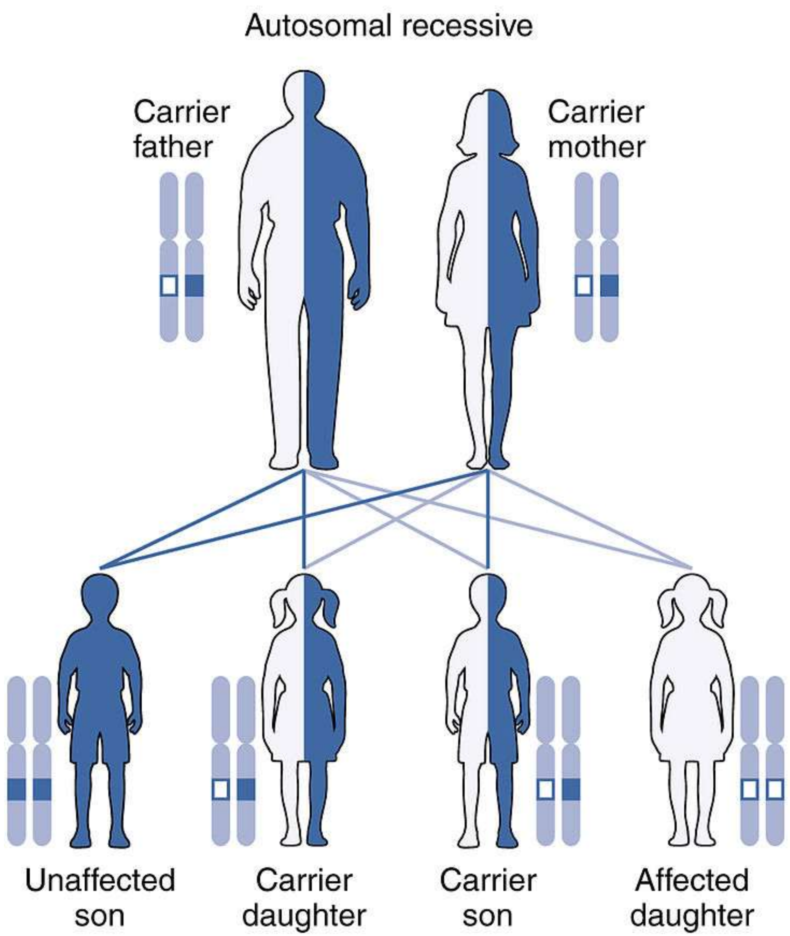
- **Genetic anomalies** are often referred to as "genetic disorders" even though in some cases they are neutral or even advantageous
  - Anomalies that are inherited can be recessive or dominant – changes the risk for offspring
    - e.g. Huntington's disease & an allele increasing the risk of breast cancer (*BRCA1*) are **dominant genetic anomalies**
      - people that are heterozygous or homozygous dominant will be affected
    - e.g. albinism is a **recessive genetic anomaly**
      - only homozygous recessive will be affected

- Recessive genetic anomalies only show up in the phenotype if the person is homozygous recessive
    - e.g. albinism = mutation that causes the pigment protein melanin to be nonfunctional – leading to little or no pigment in the eyes, hair, & skin

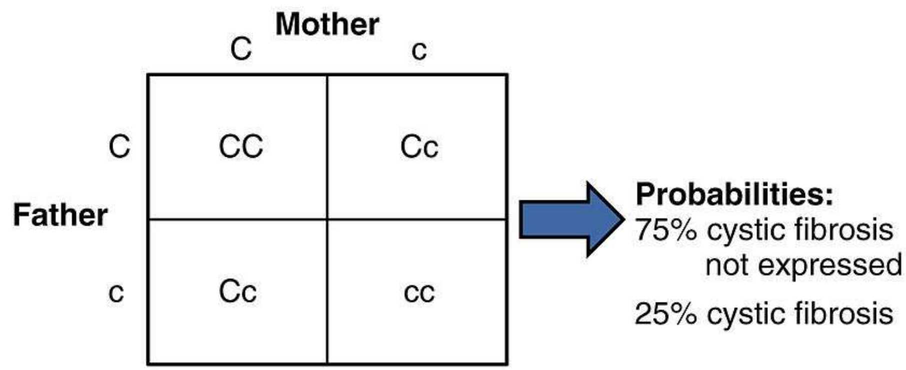
  – Both parents can be regularly pigmented but be *carriers*
    - **Carrier** = heterozygous for a regular & a recessive allele, but unaffected/not shown in their phenotype – can pass it to offspring though

Figure: https://pixabay.com/photos/kangaroo-albino-wildlife-nature-3835886/
Figure: https://commons.wikimedia.org/wiki/File:People_with_albinism.jpg

- Another example of a recessive genetic anomaly = **cystic fibrosis**: a life-threatening disorder that damages the lungs & digestive system



2 healthy parents that are **carriers** can have offspring with cystic fibrosis



The environment affects traits too

- You may have noticed before that identical twins aren't quite identical
  - This is because some traits are affected by the environment
    - Obvious traits affected by environment code for things like personality, intelligence, or body shape & weight

Figure: https://nara.getarchive.net/media/army-spc-joseph-maurino-left-and-his-identical-twin-a36683

- <u>Other traits are affected by environment too:</u>
  - e.g. people who are pregnant sometimes notice changes in their **hair texture** – this is because the environment (in this case, circulating hormones) affects gene expression of hair proteins
  - e.g. some animals have "heat-sensitive **hair color**" – the dark-pigment proteins are made from the genes but only activated under cool temperatures, like those found in the extremities (ears, tail, & paws)

Figure: https://www.pxfuel.com/en/free-photo-efkor